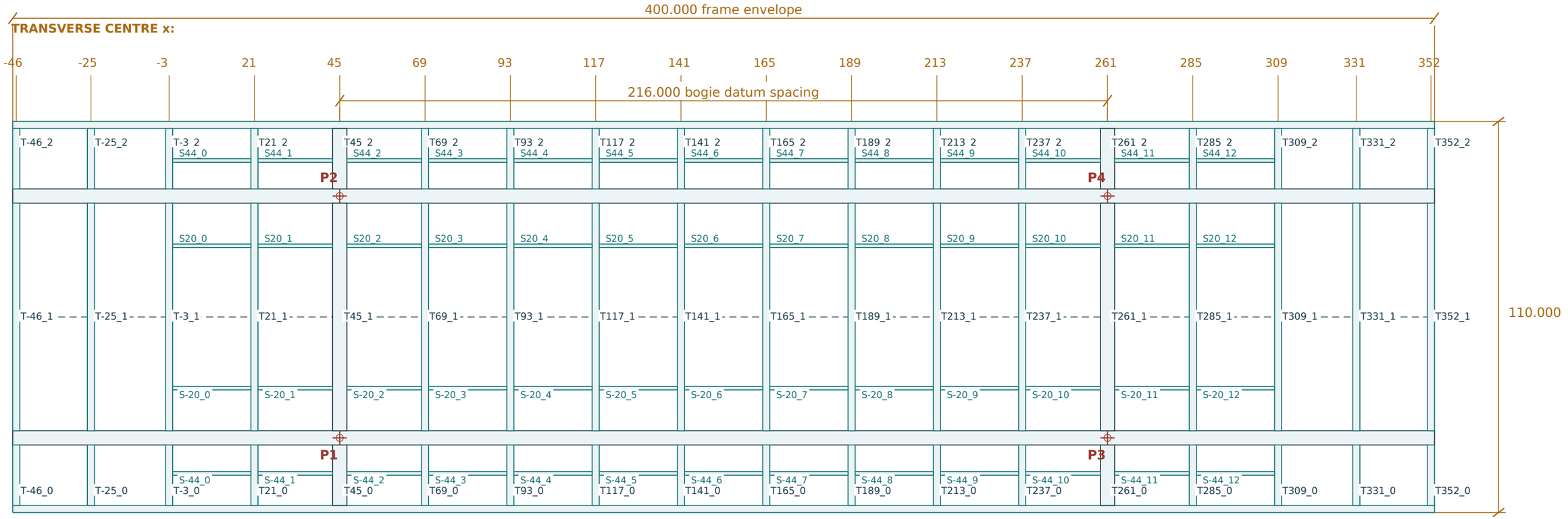
PULSIPHER FARMS / CHRISTMAS RAILWAY

FIRST COACH FRAME / DATUM AND MEMBER LOCATIONS

Retained F1: 110 steel member envelopes, 400 × 110 in overall. Dimensions locate the study geometry; fabrication release remains pending.

H1.1 TOP PLAN / ALL MEMBER IDS / DIMENSIONS IN INCHES



G-1 / G1: main girders y = −34 / +34. P-1 / P1: perimeter rails y = −54 / +54.

T station suffix: _0 negative-y arm; _1 central span; _2 positive-y arm. S rails lie at y = ±20 and ±44.

H1.2 DATUM CONTROL

A: nominal steel bearing plane z = 31.003937

B: y = 0, derived from main girder centres ±34

C: front frame face x = −47; cabin x = 0 is 47 in aft

P1–P4: measurement locations; not approved fixture contacts

H1.3 FIXED GEOMETRY / RELEASE CONDITIONS

F1 finished floor z = 32.000 study; 12 in main/bolster underside z = 19.003937. Ordinary 4 in transverse underside z = 27.003937.

Do not adjust members to suit unreleased bogie, coupler, brake, stair or floor hardware. Actual mount details and distortion tolerance need the responsible engineer’s release.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release

STEEL CUT ENVELOPES / SECTION ORIENTATION

Full grouped schedule of the retained 110 members. These are model envelopes before accepted fit-up, weld-preparation and stock allowances.

H2.1 GROUPED CUT ENVELOPES / ONE COACH

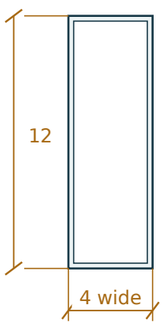
Cut group	Nominal HSS / in	Qty	Each length / in	Total / in	Bare mass / kg
CUT-G12-400	HSS 12x4x1/4	2	400.000	800.000	780.78
CUT-G12-64	HSS 12x4x1/4	2	64.000	128.000	124.93
CUT-G12-17	HSS 12x4x1/4	4	17.000	68.000	66.37
CUT-C4L-400	HSS 4x2x1/8	2	400.000	800.000	143.64
CUT-C4L-64	HSS 4x2x1/8	16	64.000	1024.000	183.86
CUT-C4L-17	HSS 4x2x1/8	32	17.000	544.000	97.67
CUT-SR-22	HSS 3x1x1/8	36	22.000	792.000	91.31
CUT-SR-21	HSS 3x1x1/8	16	21.000	336.000	38.74

TOTAL: 110 members / 1527.288 kg bare steel model. Excludes weld metal, plates, caps, mounts, coatings and equipment.

MATERIAL STUDY: ASTM A500/A500M-23 Grade C, Fy 50 / Fu 62 ksi; heat, toughness and vehicle application not accepted. Sources C3HS01–03.

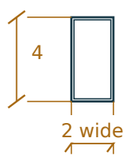
H2.2 SECTION ORIENTATION / NOMINAL WALL

G12 / HSS 12x4x1/4



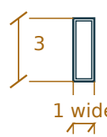
t nominal 0.25 in

C4L / HSS 4x2x1/8



t nominal 0.125 in

SR / HSS 3x1x1/8



t nominal 0.125 in

Square-line corners are a diagram convention; actual corner radii and measured wall require product data.

H2.3 MEMBER / END IDENTIFICATION

G12: G−1 / G1 at 400; T45_1 / T261_1 at 64; four bolster arms at 17.

C4L: P−1 / P1 at 400; sixteen ordinary central spans at 64; thirty-two arms at 17.

SR: each y = ±20 / ±44 rail has 13 pieces. Index 1, 2, 10, 11 are 21 in; other indices 0, 3–9, 12 are 22 in.

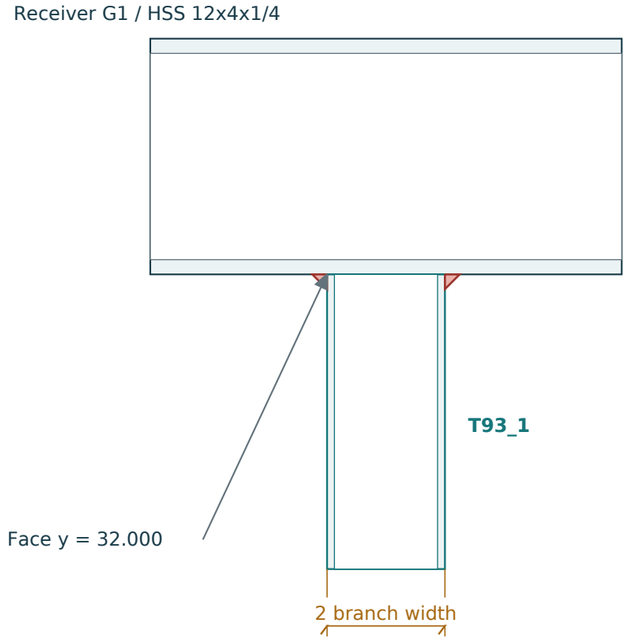
Each part retains its H01 ID. The current envelope ends on the receiver face. Manufacturing length must incorporate the accepted end preparation and fit-up gap; no saw tolerance, bevel, root gap or weld shrinkage allowance is released by this table.

ORDINARY CROSSMEMBER JOINT / CM-01

WELD-METAL-ONLY CALCULATION TRIAL — NOT A SELECTED JOINT OR WPS. Local HSS wall, fatigue, heat input and complete connection remain unresolved.

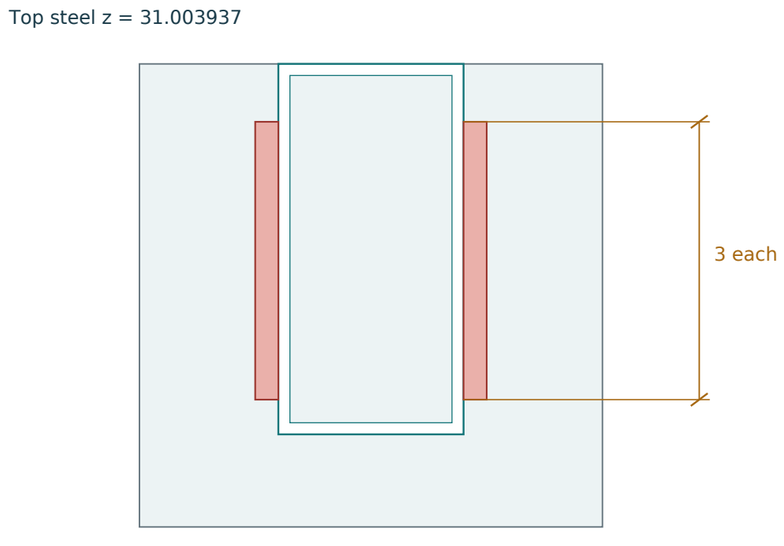
H3.1 PLAN SECTION / ASSEMBLED ENVELOPES

J-CM:T93_1:end



Sharp-line HSS corners are schematic; actual section/profile required.

H3.2 END ELIVATION / TWO SIDE FILLET TRIALS

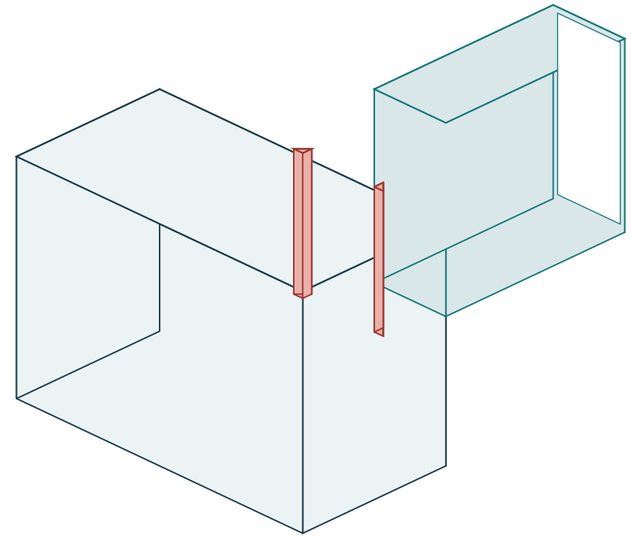


Weld z = 27.378937 to 30.378937 in

Leg 0.25 in / lower offset 0.375 in

No top-surface weld in this trial. Receiver view cropped.

H3.3 EXPLODED RELATIONSHIP



Members shortened and separated for view only.

Red prisms = calculated trial weld envelopes.

No plates, bolts or holes added to complete the picture.

H3.4 STATED DEMAND / FULL-RESTRAINT SENSITIVITY

V = 5.860 kN; end M = 1.588 kNm; receiver-axis T = 0.298 kNm

4.8 kPa floor live + 1.0 kPa dead; 24 in tributary width; 64 in face span. Two-times gravity sensitivity is not an adopted vehicle load factor.

Trial leg exceeds the 1/8 in nominal branch wall. Local wall plastification/punching, root eccentricity, effective length, weldability, fatigue and WPS acceptance are not resolved. Do not generalize this example to its entire joint family.

H3.5 WELD-METAL COMPARISON RATIOS / NOT JOINT CAPACITY

Leg / in	0% restraint	50%	100%
0.125	0.079	0.845	1.684
0.1875	0.053	0.563	1.123
0.25	0.039	0.422	0.842

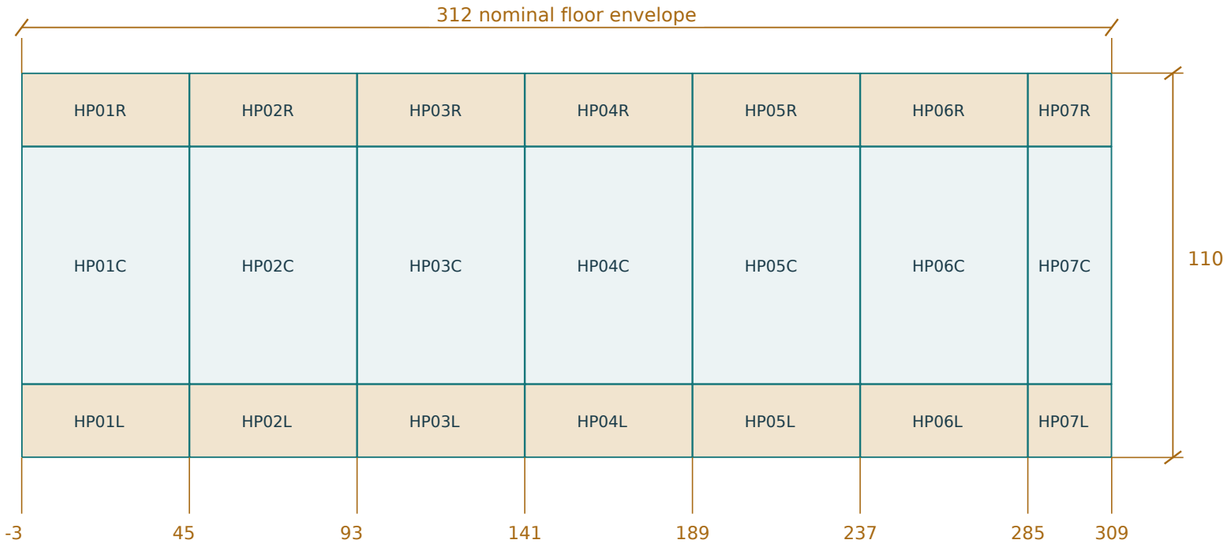
Ratio compares calculated weld-metal stress to 217.185 MPa assumed comparison strength (E70, φ=.75, 0.60FEXX; no directional increase). A ratio below 1 does not accept the full HSS connection.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release

FLOOR PANELS / CUT ENVELOPES / SUPPORTED SEAMS

H-P125 is a separate fit-up study: 1/8 in internal gap, 1/16 in perimeter inset. Panel edge closures, bonds and attachment details remain unreleased.

H4.1 H-P125 PANEL PLAN / 21 SUPPLIER-FINISHED ENVELOPES



Longitudinal seams y = ±34 over main girders; seam coverage is nominal outer-envelope only.

H4.2 FINISHED PANEL ENVELOPES / INCHES

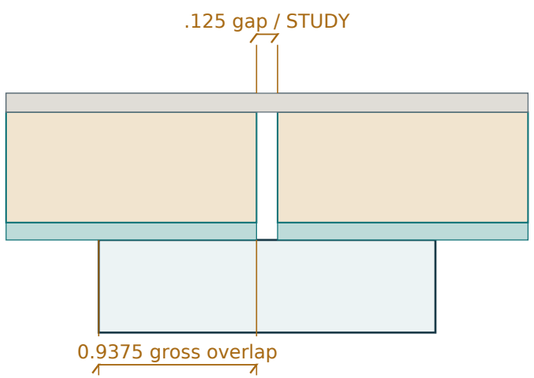
IDs	Qty	Length × width × depth
HP01–06 L/R	12	47.875 × 20.875 × .750
HP01–06 C	6	47.875 × 67.875 × .750
HP07 L/R	2	23.875 × 20.875 × .750
HP07 C	1	23.875 × 67.875 × .750

Candidate: Plascore AA3.6-80, 0.75 in nominal panel. Dimensions describe the finished closed-edge envelope, not raw honeycomb-core cuts. Factory edge sections, skins, grain direction, trims and machining allowances must be returned.

Catalog top-flat sensitivity leaves 0.6565 in/panel at the 2 in support and 1.3750 in/panel at the 4 in support. Actual usable contact, pressure and bearing acceptance remain unresolved (H-B09).

H4.3 TRANSVERSE SEAM x = 93

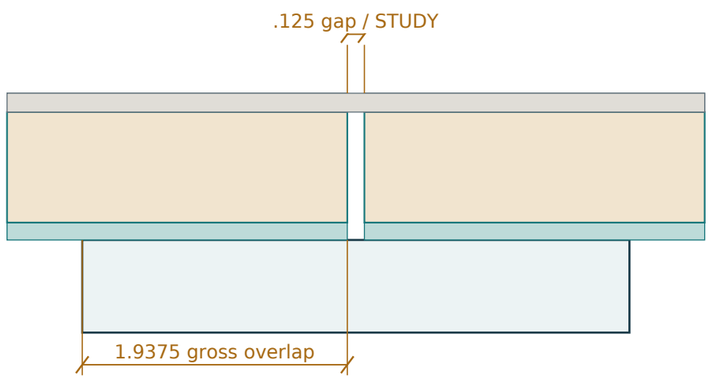
Finish shown continuous; joint treatment HELD.



T93_1 HSS 4×2×1/8 / shown depth broken

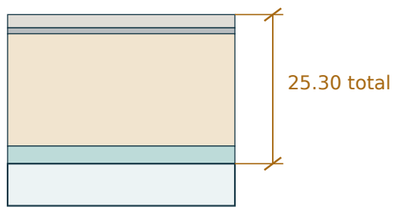
H4.4 LONGITUDINAL SEAM y = 34

Finish shown continuous; joint treatment HELD.



G1 HSS 12×4×1/4 / shown depth broken

H4.5 FLOOR LAYER STACK / mm



2.25 finish + 1.00 adhesive reserve

19.05 panel + 3.00 support reserve

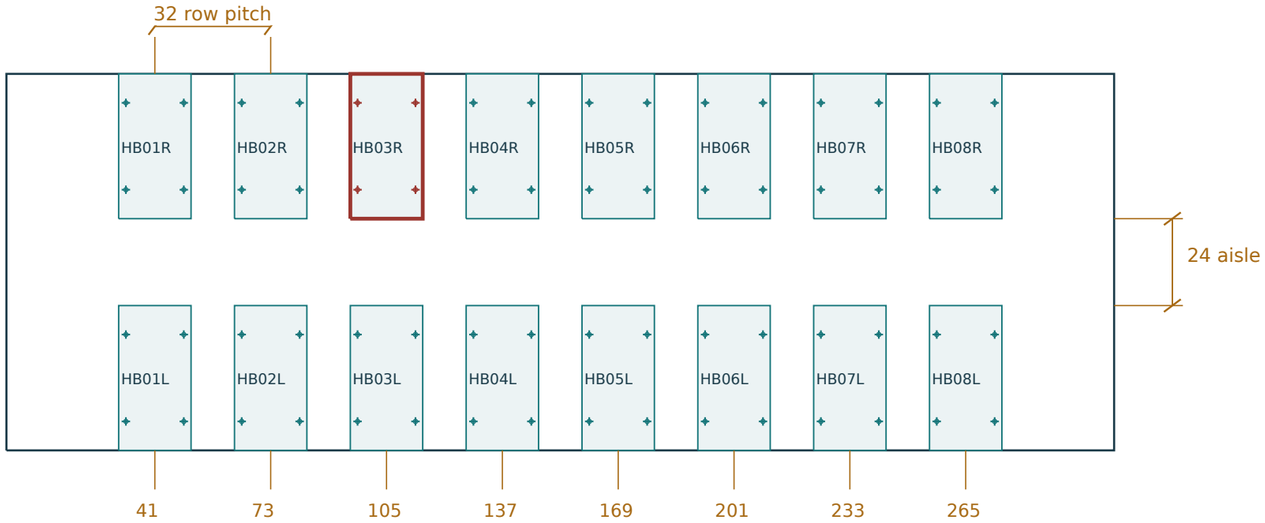
Gross overlap uses rectangular envelopes; rounded corners and surface condition reduce actual contact. Contact, bond, sealing and bearing strength remain unresolved.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release

REGULAR COACH BENCHES / HARDPOINT LOAD PATH

H-S16 is a mounting study, not a selected seat or anchor. 16 bench envelopes and 64 hardpoint axes use the retained F1 steel without moving members.

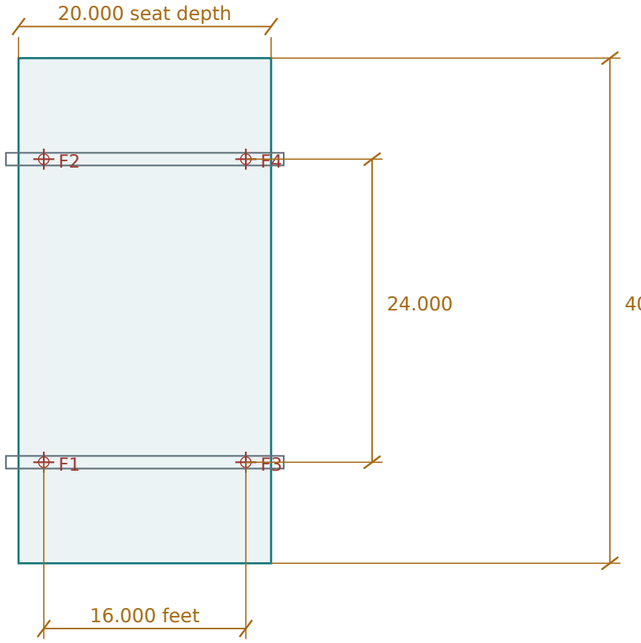
H5.1 REGULAR COACH PLAN / H-S16 HARDPOINT AXES



Row centres x / in. Bench seat envelope = 20 deep × 40 wide; seat top = 18 above floor.

Eight former hardpoint centres landed at member faces. H-S16 changes longitudinal foot spacing to 16 in; all 64 centres lie inside existing rail spans. This alone does not establish anchor capacity.

H5.2 HB03R / PROJECTED STEEL AND FOOT CENTRES



x feet: 97 / 113

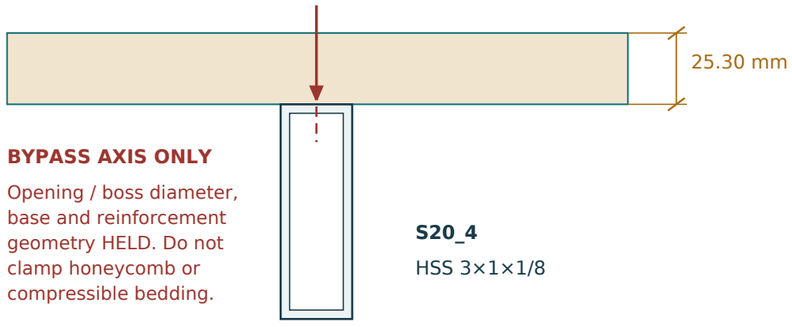
y feet: 20 / 44

S20_4 and S44_4 run x94..116. Axes only; no bolt-hole sizes.

Rail 1 in overall; catalog workable top flat only .438 in. Local hardpoint design HELD.

Example H-S1 is at x97 / y20; SR-01 on H07 is a separate row73 joint example.

H5.3 H-S1 LOAD-BYPASS SECTION



25.30 mm is datum separation; subtract any above-steel reinforcement.

H5.4 EXPLODED COMPONENT ROLES

BENCH BASE / REMOVABLE FASTENER
Actual base, bolt, torque and wrench access HELD

↓

LOAD BYPASS / SEALED OPENING
Axis only; skin / seal / bedding carry no anchor load

↓

ENGINEERED RAIL HARDPOINT
Cap/saddle/internal reinforcement dimensions HELD

H5.5 PER-FOOT DEMAND / kN

Case	Comp.	Uplift	Shear
Gravity	0.748	0.000	0.000
Long 1g	2.991	1.496	0.748
Lat .5g	1.496	0.000	0.374

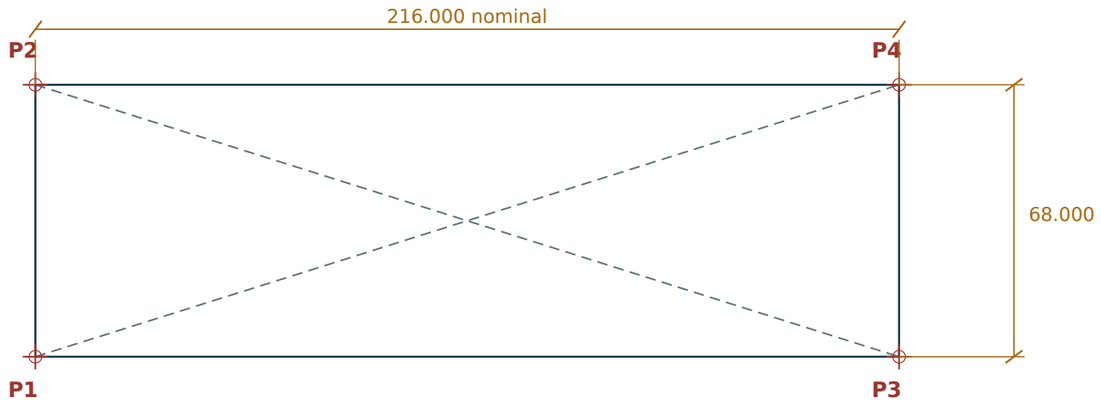
305 kg loaded bench; assumed CG 24 in above floor. Separate axis sensitivities, not a governing seat-restraint standard or accepted anchor demand set.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release

FIRST-ARTICLE FABRICATION / DIMENSIONAL RECORD

A build traveler and measurement scheme for engineering development. Stock-tube tolerances do not establish finished-vehicle tolerances.

H6.1 DATUM TRANSFER / FOUR REFERENCE LOCATIONS



P1–P4 and P2–P3 diagonal = 226.450878 in (calculated)

All four steel-top targets z = 31.003937 in; fixture contacts are not assigned.

H6.2 TRAVELER MEASUREMENT FIELDS

Feature / in	Nominal	Allowed	Measured
Main span / G centre	68.000	_____	_____
Bogie x spacing	216.000	_____	_____
P1–P4 diagonal	226.450878	_____	_____
P2–P3 diagonal	226.450878	_____	_____
A-plane profile	Engineer map	_____	_____
Joint fit-up gap	Joint drawing	_____	_____

Allowed values must be released before cutting or fit-up. Record instrument ID, calibration, temperature, support condition, inspector and disposition for each measurement set.

H6.3 CONTROLLED FABRICATION SEQUENCE / RETAIN RECORDS

H-F01	Match 110 part IDs, section/material candidates and joint revisions.
H-F02	Record heat/certificate, wall, bow, twist, seam and damage; quarantine mismatch.
H-F03	Survey A/B/C datums with calibrated instruments and released fixture plan.
H-F04	Convert model length to released part length; label heat, orientation and ends.
H-F05	Dry-fit main girders, separate bolster segments and perimeter rails.
H-F06	Fit transverse and seat rails; verify top plane, seam bearing and tool access.
H-F07	Log joint ID, WPS, welder, distortion and required inspection before concealment.
H-F08	Trial-fit panel edges and seat hardpoints; verify direct load transfer to steel.
H-F09	After inspection, apply accepted coating, edge seal and isolation systems.
H-F10	Weigh and survey first article; record deviations and authorized next tests.

Farm crew: labels, records, handling, supervised measurements and released finish work. Specialist: fixture/joint design, qualified welding, NDT/inspection and acceptance. No occupied test or production duplication is released.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release

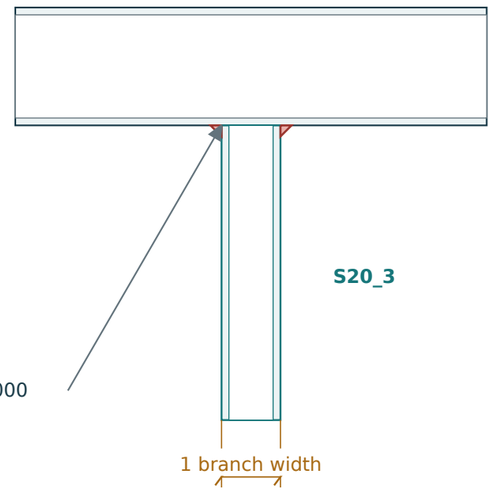
SEAT-RAIL END JOINT / SR-01

WELD-METAL-ONLY CALCULATION TRIAL — NOT A SELECTED JOINT OR WPS. Local HSS wall, fatigue, heat input and complete connection remain unresolved.

H7.1 PLAN SECTION / ASSEMBLED ENVELOPES

J-SC:S20_3:start

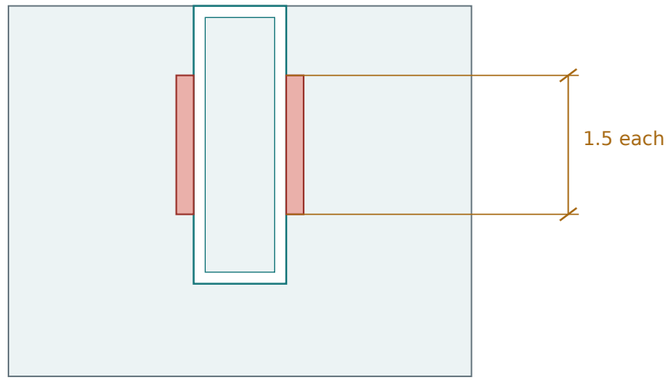
Receiver T69_1 / HSS 4x2x1/8



Sharp-line HSS corners are schematic; actual section/profile required.

H7.2 END ELEVATION / TWO SIDE FILLET TRIALS

Top steel z = 31.003937

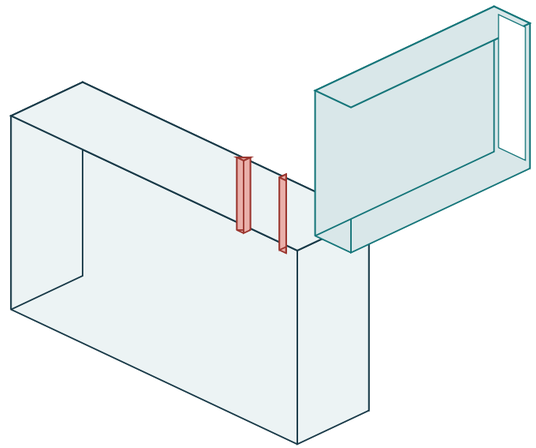


Weld z = 28.753937 to 30.253937 in

Leg 0.1875 in / lower offset 0.75 in

No top-surface weld in this trial. Receiver view cropped.

H7.3 EXPLODED RELATIONSHIP



Members shortened and separated for view only.

Red prisms = calculated trial weld envelopes.

No plates, bolts or holes added to complete the picture.

H7.4 STATED DEMAND / FULL-RESTRAINT SENSITIVITY

V = 0.773 kN; end M = 0.107 kNm; receiver-axis T = 0.020 kNm

305 kg loaded bench; H-S16 foot x81/y20 on S20_3, face span x70..92. Two-times gravity sensitivity; actual anchor / abuse / crash criteria remain separate.

Trial leg exceeds the 1/8 in nominal branch wall. Local wall plastification/punching, root eccentricity, effective length, weldability, fatigue and WPS acceptance are not resolved. Do not generalize this example to its entire joint family.

H7.5 WELD-METAL COMPARISON RATIOS / NOT JOINT CAPACITY

Leg / in	0% restraint	50%	100%
0.125	0.021	0.227	0.453
0.1875	0.014	0.151	0.302
0.25	0.010	0.114	0.227

Ratio compares calculated weld-metal stress to 217.185 MPa assumed comparison strength (E70, φ=.75, 0.60FEXX; no directional increase). A ratio below 1 does not accept the full HSS connection.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release